Exp_10.l

```
%{
#include "y.tab.h"
%}

%%
"for"           { return FOR; }
[a-zA-Z_][a-zA-Z0-9_]*  { return ID; }
[0-9]+          { return NUM; }
"="             { return '='; }
"<"             { return '<'; }
"++"            { return INC; }
";"             { return ';'; }
"("             { return '('; }
")"             { return ')'; }
"{"             { return '{'; }
"}"             { return '}'; }
"+"             { return '+'; }
"-"             { return '-'; }
[ \t\n]+        ;
.               { return yytext[0]; }
%%

int yywrap()
{
    return 1;
}
```

Exp_10.l

```
%{
#include <stdio.h>
#include <stdlib.h>

int yylex();
void yyerror(const char *s);
%}

%token FOR ID NUM INC

%%
stmt:
    FOR '(' init ';' cond ';' update ')' '{' '}'
    {
        printf("Valid for loop\n");
    }
    ;

init:
    ID '=' NUM
    ;

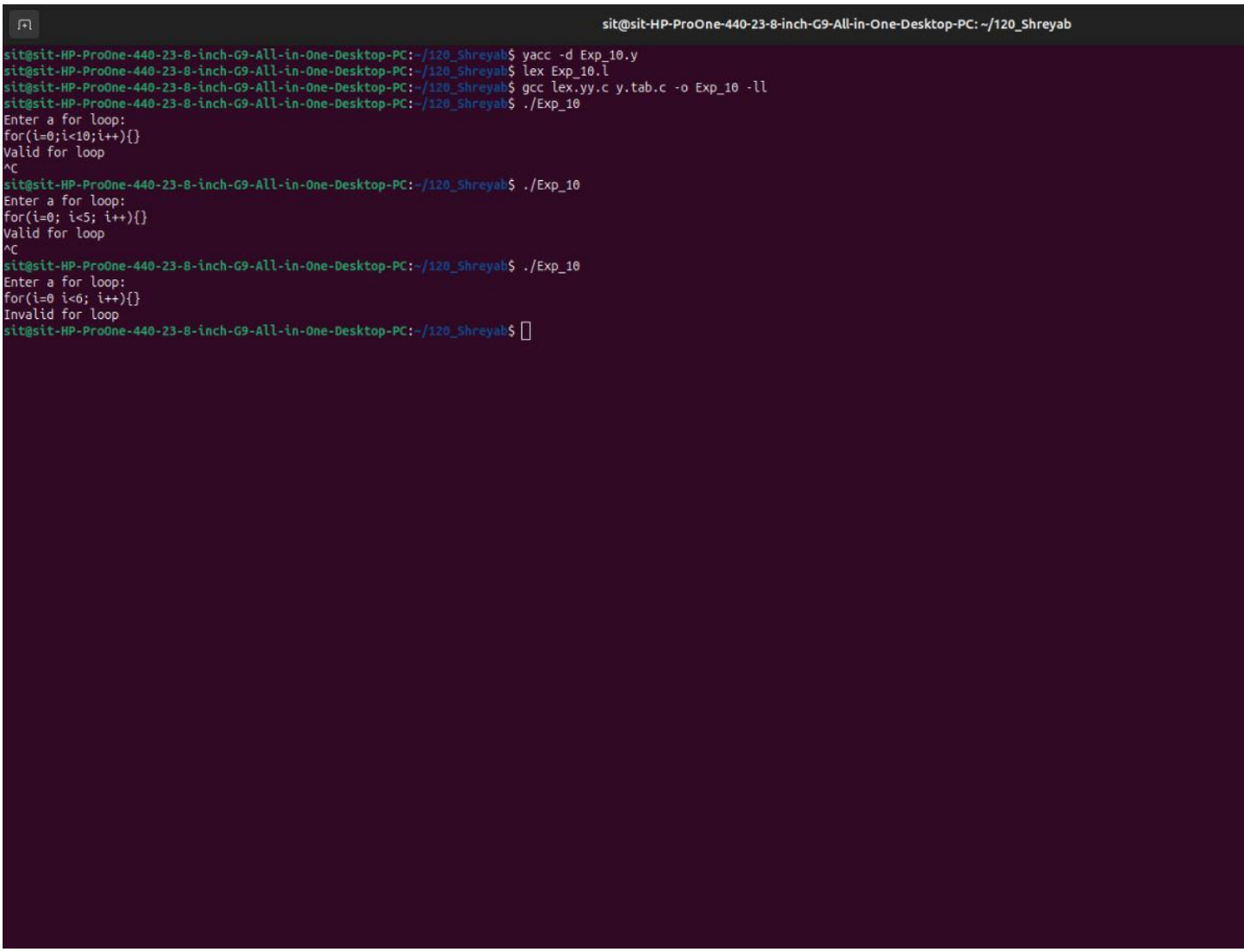
cond:
    ID '<' NUM
    ;

update:
    ID INC
    | INC ID
    ;

%%

void yyerror(const char *s)
{
    printf("Invalid for loop\n");
}

int main()
{
    printf("Enter a for loop:\n");
    yyparse();
```


sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC: ~/120_Shreyab
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/120_Shreyab$ yacc -d Exp_10.y
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/120_Shreyab$ lex Exp_10.l
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/120_Shreyab$ gcc lex.yy.c y.tab.c -o Exp_10 -ll
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/120_Shreyab$ ./Exp_10
Enter a for loop:
for(i=0;i<10;i++){}
Valid for loop
^C
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/120_Shreyab$ ./Exp_10
Enter a for loop:
for(i=0; i<5; i++){}
Valid for loop
^C
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/120_Shreyab$ ./Exp_10
Enter a for loop:
for(i=0 i<6; i++){}
Invalid for loop
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/120_Shreyab$